MISSION ENGINEERING · PRIMER

Measure the mission, not the activity.

Mission engineering starts from what the mission must achieve and works down to the systems that support it. Evidence then rolls back up, so every decision is judged by its effect on the mission.

What it is

The deliberate planning, analysis, and integration of operational and system capabilities to achieve the mission outcomes we need.

Top down, mission need is decomposed: mission, then the tasks that accomplish it, then the capabilities those tasks require, then the systems that deliver them.

Bottom up, evidence from instrumentation rolls back through the same chain, along the mission thread: the end-to-end sequence of tasks that gets the mission done.

Why it matters

Activity is not outcome. Deployments, features, and closed tickets can all rise while the mission falls behind.

Traceability. Every service rolls up to a capability, a task, and the mission, so removing one names exactly which measure falls.

One shared language. Mission owners, tradecraft experts, designers, and engineers judge the same mission, each at their own level.

Honest thresholds. We baseline first, then mission analysis sets what "required" means at each level.

Four levels, four measures

Each level answers one question

MISSION NEED FLOWS DOWN ↓

OPERATIONAL Frames the mission	**Measure of Success** Objectives met vs. expected. *Is the mission succeeding?*	**In our strategy:** mission owners assess it, informed by the evidence below, never generated by it
FUNCTIONAL Structures the tasks	**Measure of Effectiveness** Conditions achieved vs. required. *Can people accomplish the mission's tasks?*	**Stratos:** task success on mission workflows the triad defines
CAPABILITY Aligns the capabilities	**Measure of Capability** Sufficiency vs. required. *Are our capabilities enough for those tasks?*	**MACH1:** SLO attainment and workflow health per capability platform
PHYSICAL Implements the systems	**Measure of Performance** Performance vs. expected. *Do the systems perform as specified?*	**MACH1:** availability, errors, and latency against published SLOs

ALONGSIDE — **Mission Control:** delivery sits beside the four levels, not in them. It measures how fast the organization restores any measure that falls.

↑ EVIDENCE ROLLS UP

How it enables better outcomes

1
Prioritize by mission impact
Work is ranked by the measure it moves, not by who asks loudest.

2
Design from mission tasks
Workflows come from the triad's decomposition of the mission, so task success truly measures effectiveness.

3
Invest where capability falls short
Capability measures show which platforms cannot yet carry the tasks that depend on them.

4
Catch and restore fast
Deployment annotations tie a falling measure to its cause, and delivery speed sets how fast it recovers.

5
Keep judgment human
Mission owners assess success. Telemetry informs the call but never makes it.

Traceability in practice: one chain, read both ways

Bottom up it explains what happened; top down it shows which fix restores the mission

DELIVERY
A release ships to production
→
PERFORMANCE
Client errors jump on one endpoint
→
CAPABILITY
The capability behind a task drops below required
→
EFFECTIVENESS
Analysts abandon a mission task; support contacts spike
→
SUCCESS
Mission owners see an objective at risk, and why